Statistics Hybrid Practice Set – S15 (L11)

Main Focus

CV + Box Plot + Distribution Interpretation

This set gradually introduces **Coefficient of Variation (CV)** while continuing to reinforce quartiles, IQR, outliers, and box plots.

Q1. Compare Consistency Using Standard Deviation

Two machines produce parts with these measurements:

Machine A: 98, 99, 100, 101, 102
Machine B: 80, 90, 100, 110, 120

Answer:

1. Which machine has the smaller range?
2. Which machine appears more consistent?
3. Why is standard deviation useful for comparing consistency?

Q2. Coefficient of Variation — New ⭐

Two stores have different average daily sales:

Store	Mean Sales	Standard Deviation
A	₹50,000	₹5,000
B	₹80,000	₹8,000

Calculate the **CV for both stores**.

Then answer:

1. Which store has the lower relative variation?
2. Which store is more consistent relative to its average sales?

Q3. CV Interpretation

Two investment options have:

Investment	Mean Return	Standard Deviation
A	10%	2%
B	20%	5%

Calculate CV for both.

Then determine:

Which investment has lower relative variability?

Don't judge consistency using standard deviation alone.

Q4. Five Number Summary

The following dataset represents employee experience in years:

1, 2, 3, 4, 5, 6, 7, 8, 12

Find:

1. Minimum
2. Q1
3. Median
4. Q3
5. Maximum
6. IQR

Then create the **Five Number Summary**.

Q5. Outlier Detection

Using the same dataset from Q4:

1, 2, 3, 4, 5, 6, 7, 8, 12

Calculate:

1. Lower Fence
2. Upper Fence
3. Is 12 an outlier?
4. If yes, how would it appear on a box plot?

Q6. Box Plot Interpretation

A company's box plot has:

- Minimum = 10
- Q1 = 20
- Median = 25
- Q3 = 40
- Maximum = 45

Answer:

1. IQR?
2. What percentage of observations lie between 20 and 40?
3. Is the median closer to Q1 or Q3?
4. Which half of the middle 50% has greater spread?
5. What does this suggest about the distribution?

Q7. Compare Box Plots

Two departments have these Five Number Summaries:

Statistic	Dept A	Dept B
Minimum	20	10
Q1	30	20
Median	40	35
Q3	50	60
Maximum	60	100

Answer:

1. Which department has the higher median?
2. Which has the larger IQR?
3. Which has the larger overall range?
4. Which department appears more variable?
5. Which department would you prefer if you wanted more consistent values?

Explain your reasoning.

Q8. Distribution Shape

A box plot has:

- Q1 = 20
- Median = 22
- Q3 = 35
- Upper whisker = 60
- Lower whisker = 15

Answer:

1. Is the median closer to Q1 or Q3?
2. Which side has the longer whisker?
3. Does the distribution appear more likely to be left-skewed or right-skewed?
4. What feature of the box plot supports your answer?

Q9. Data Analyst Scenario

A Data Analyst needs to compare the **relative consistency of salaries** between two departments, but the departments have very different average salaries.

Which measure is more appropriate?

A. Range
B. Variance
C. Coefficient of Variation
D. Median

Explain why.

Q10. Mixed Business Case

Two delivery companies have the following statistics:

Company	Mean Delivery Time	Standard Deviation
A	30 min	3 min
B	50 min	4 min

Answer:

1. Calculate CV for both companies.
2. Which company has lower relative variation?
3. Which company is more consistent relative to its average delivery time?
4. If the company wants **consistency**, which one would you choose?
5. Why would looking only at standard deviation give an incomplete comparison?